THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 7 — SCIENCE · NCERT-TYPE CHAPTER TEST SET 3

Chapter 6: **Heat Transfer in Nature** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The vacuum between double walls of a flask prevents: 1

(a) sound **(b)** light
(c) radiation only **(d)** conduction and convection

2. Which of these is a poor conductor (insulator) of heat? 1

(a) silver **(b)** iron **(c)** wood **(d)** copper

3. The land breeze blows during the: 1

(a) noon **(b)** day **(c)** night **(d)** morning

4. The cork in a thermos flask is used because it: 1

(a) is shiny **(b)** is heavy
(c) is a poor conductor **(d)** is a good conductor

5. A steel spoon dipped in hot tea gets hot at the other end. This is due to: 1

(a) evaporation **(b)** convection **(c)** radiation **(d)** conduction

6. During the day the land gets hot faster than the sea, so the air above land: 1

(a) sinks **(b)** stays **(c)** freezes **(d)** rises

7. A bright surface is a good ___ of heat radiation. 1

(a) reflector **(b)** absorber **(c)** emitter **(d)** conductor

8. Warm air above a heater rises. This is due to: 1

(a) conduction **(b)** convection **(c)** condensation **(d)** radiation

9. Sea breeze blows from the sea towards the land because: 1

(a) sea is hot **(b)** wind from mountains
(c) air over land is warmer and rises **(d)** land is cold

10. A dark rough surface is a good ___ of heat. 1

(a) conductor **(b)** insulator **(c)** reflector **(d)** absorber

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Why is a thermos flask double-walled with vacuum? 2

12. What is radiation? Give an example. 2

13. What is a land breeze? 2

14. Why does an AC placed at the top of a room cool the whole room? 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain land breeze at night. 3

16. Why does a pillow or quilt keep us warm? 3

17. Explain why the metal tip of a soldering iron has a wooden handle. 3

OR

Why do we feel cooler wearing white clothes in summer? 3

18. Why is water heated from below and not from above? 3

OR

Why are igloos warm inside? 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain how a thermos flask minimises all three modes of heat transfer. 5

OR

Explain why we use fans in summer and not in cold weather and how the air movement helps. 5

20. Give reasons: 5

(a) pans have wooden handles,
(b) the freezer is at the top of a fridge,
(c) clothes in summer are light-coloured,
(d) a blanket traps heat,
(e) the ice box is covered.

OR

Describe how a solar cooker uses reflectors, glass and black surface. 5

END OF QUESTION PAPER — Total: 40 Marks

MASTERMIND